

Part B — Data Exploration & Analysis

1. Introduction & Dataset Overview

This report explores the 2024 CMS Medicare Physician & Other Practitioners - by Provider and Service dataset. The dataset contains information on Medicare services, utilization, and payments for healthcare providers. The goal of this analysis is to understand the data and prepare it for unsupervised anomaly detection of unusual billing patterns.

2. Data Loading & Initial Inspection

The raw dataset contains 9,781,673 rows and 28 columns. Each row represents a combination of a provider (NPI), HCPCS service code, and place of service, rather than a single provider.

There are 1,207,473 unique providers, 6,471 HCPCS codes, and 113 provider types. The columns include provider information, service information, utilization, and payment amounts.

ID-like columns such as NPI, ZIP, FIPS, HCPCS, and RUCA were loaded as strings to preserve leading zeros. Low-cardinality text columns were converted to categorical variables.

There were no missing values in the numeric columns. Some providers had foreign addresses and were removed during data cleaning.

	dtype	n_unique	example
Rndrng_NPI	object	1207473	1003000126
Rndrng_Prvrdr_Last_Org_Name	object	290816	Enkeshafi
Rndrng_Prvrdr_First_Name	object	84734	Ardalan
Rndrng_Prvrdr_MI	category	32	nan
Rndrng_Prvrdr_Crdntls	object	18018	M.D.
Rndrng_Prvrdr_Ent_Cd	category	2	I
Rndrng_Prvrdr_St1	object	425655	6410 Rockledge Dr Ste 304
Rndrng_Prvrdr_St2	object	51376	None
Rndrng_Prvrdr_City	object	12736	Bethesda
Rndrng_Prvrdr_State_Abrvtn	category	62	MD
Rndrng_Prvrdr_State_FIPS	category	61	24
Rndrng_Prvrdr_Zip5	object	20746	20817
Rndrng_Prvrdr_RUCA	category	22	1
Rndrng_Prvrdr_RUCA_Desc	category	15	Metropolitan area core: primar
Rndrng_Prvrdr_Cntry	category	25	US
Rndrng_Prvrdr_Type	category	113	Internal Medicine
Rndrng_Prvrdr_Mdcr_Prtcptg_Ind	category	2	Y
HCPCS_Cd	object	6471	99221
HCPCS_Desc	object	6462	Initial hospital care with str
HCPCS_Drug_Ind	category	2	N
Place_Of_Srvcs	category	2	F
Tot_Benes	int64	9361	36
Tot_Srvcs	float64	31765	36.0
Tot_Bene_Day_Srvcs	int64	11752	36
Avg_Sbmtld_Chrg	float64	2197498	284.37888889
Avg_Mdcr_Alowd_Amt	float64	6570314	79.063888889
Avg_Mdcr_Pymt_Amt	float64	6332463	60.828888889
Avg_Mdcr_Stdzd_Amt	float64	5766618	62.376388889

Fig 1. Column overview: data type, number of unique values, and an example value for each of the 28 columns.

3. Column Descriptions

- **Rndrng_NPI**: National Provider Identifier, a unique 10-digit ID of the provider
- **Rndrng_Prvrdr_Last_Org_Name**: provider's last name if an individual, or organization name if an organization
- **Rndrng_Prvrdr_First_Name**: provider's first name if an individual, blank for an organization
- **Rndrng_Prvrdr_MI**: provider's middle initial
- **Rndrng_Prvrdr_Crdntls**: credentials of the provider, such as MD, DO, NP, etc.
- **Rndrng_Prvrdr_Ent_Cd**: whether the provider is an individual (I) or an organization (O)
- **Rndrng_Prvrdr_St1 / St2**: provider's street address, lines 1 and 2
- **Rndrng_Prvrdr_City**: provider's city

- **Rndrng_Prvrdr_State_Abrvtn**: provider's state/territory
- **Rndrng_Prvrdr_State_FIPS**: standard numeric code for the state/territory
- **Rndrng_Prvrdr_Zip5**: provider's 5-digit ZIP code
- **Rndrng_Prvrdr_RUCA**: Rural-Urban Commuting Area code (urban vs. rural classification of the ZIP code)
- **Rndrng_Prvrdr_RUCA_Desc**: human-readable RUCA description
- **Rndrng_Prvrdr_Cntry**: provider's country code
- **Rndrng_Prvrdr_Type**: provider specialty/type, derived from the specialty with the largest number of services
- **Rndrng_Prvrdr_Mdcr_Prtcptg_Ind**: whether the provider participates in Medicare and accepts the Medicare allowed amount as full payment (Y/N)
- **HCPCS_Cd**: code identifying the medical procedure, service or product
- **HCPCS_Desc**: plain-English description of the HCPCS code
- **HCPCS_Drug_Ind**: whether the HCPCS code is a Medicare Part B drug (Y/N)
- **Place_Of_Srvc**: F = facility (e.g. hospital), O = non-facility (office-like setting)
- **Tot_Benes**: number of distinct Medicare beneficiaries who received this service for this NPI + HCPCS + place of service combination
- **Tot_Srvcs**: number of services provided; the unit depends on the code (visits/procedures, drug dose units or miles, see Section 4.4)
- **Tot_Bene_Day_Srvcs**: number of distinct beneficiary-days of service; counts one patient on one day only once, even if several units were given
- **Avg_Sbmtd_Chrg**: average amount the provider charged (billed) for the service
- **Avg_Mdcr_Alowd_Amt**: average amount Medicare allows: Medicare payment + patient deductible/coinsurance + any third-party payment
- **Avg_Mdcr_Pymt_Amt**: average amount Medicare actually paid after deductible and coinsurance
- **Avg_Mdcr_Stdzd_Amt**: Medicare payment with geographic price differences removed, so it can be compared across regions

4. Data Quality & Cleaning

4.1 Missing values

Most values are not missing, though there are a couple of columns with 35%, and 78% of data missing values

Column	Missing %
Rndrng_Prvrdr_St2	78.08
Rndrng_Prvrdr_MI	35.42

Rndrng_Privr_Crdntls	11.04
Rndrng_Privr_First_Name	4.85
Rndrng_Privr_RUCA / RUCA_Desc	0.05
Rndrng_Privr_State_FIPS	<0.01
Rndrng_Privr_Zip5	<0.01

Table 1. Columns with missing values (all other columns are complete).

Missing values are structural, not errors: MI and St2 are optional fields, First_Name and Credentials are blank for organizations, and RUCA is missing for only 0.05% of rows. None of the key numeric variables have missing values.

4.2 Duplicates

```
1 print("Exact duplicate rows:", df.duplicated().sum())
✓ [10] 19s 757ms

Exact duplicate rows: 0
```

Fig 2. Exact duplicate check.

There are no exact duplicate rows. There are also no duplicate NPI $\times$ HCPCS $\times$ place of service combinations, so this combination is a valid unique key.

4.3 Logical consistency checks

Duplicates are not the only possible problem, so we also checked that the values make sense given what each column means:

- $\text{Tot_Benes} \leq \text{Tot_Bene_Day_Srvc}$ – every patient was seen on at least one day
- $\text{Tot_Bene_Day_Srvc} \leq \text{Tot_Srvc}$ – every patient-day includes at least one service
- $\text{Avg_Mdc_Pymt_Amt} \leq \text{Avg_Mdc_Alowd_Amt}$ – the allowed amount is Medicare's payment plus the patient's share, so Medicare cannot pay more than allowed

	n_rows	pct
duplicate NPI x HCPCS x POS	0	0.0000
Srvcs < Benes	34	0.0003
Bene_Days > Srvcs	72	0.0007
Bene_Days < Benes	0	0.0000
non-integer Srvcs (non-drug)	8298	0.0848
Pymt > Alowd	0	0.0000
Alowd == 0	18	0.0002
Pymt == 0	94	0.0010
Sbmtld >= 99,999 (possible cap)	28	0.0003

Table 2. Results of logical checks

- Payment never exceeds the allowed amount, and bene-days never fall below patients, which confirms our interpretation of the columns.
- 90 rows break volume logic or have a \$0 allowed amount (which makes the charge/allowed ratio undefined), so they were removed.
- \$0 payments (e.g. the patient or another insurer covered the full amount) and charges $\geq$ \$99,999 (possibly a field cap) are unusual but not errors, so we kept them.
- Fractional service counts for non-drug codes were investigated further (Section 4.4).

4.4 Units of Tot_Srvcs

HCPCS_Cd	HCPCS_Desc
A0425	Ground mileage, per statute mile
7668	
A0436	Rotary wing air mileage, per statute mile
308	
P9603	Travel allowance one way in connection with medically necessary laboratory specimen collection drawn from home bound or nursing home bound patient; prorated miles actually traveled
70	
A0435	Fixed wing air mileage, per statute mile
62	
88341	Special stained specimen slides to examine tissue, each additional procedure
14	
96372	Injection of drug or substance under skin or into muscle
10	
A0428	Ambulance service, basic life support, non-emergency transport, (bls)
9	
99213	Established patient office or other outpatient visit with low level of decision making, if using time, 20 minutes or more
8	
77067	Screening mammography
8	
90837	Psychotherapy, 1 hour
7	

Fig 3. Non-drug codes with fractional service counts.

Tot_Srvcs does not always count visits or procedures. The unit depends on the HCPCS code:

- **Drug codes** (HCPCS_Drug_Ind = Y) count dose units (e.g. per 1 mg or 0.1 mg). This explains why some drug rows have fractional services or fewer services than patients, and why 24,806 rows have charges under \$1: 97% of them are drugs priced per tiny unit, with allowed amounts equally small (median \$0.13), so these values are valid.
- **Transport codes** count miles. 8,298 non-drug rows had fractional service counts; almost all of them are mileage codes: A0425 (ground ambulance), A0435 / A0436 (air ambulance) and P9603 (lab specimen travel).
- **All other codes** count visits / procedures. We added a segment column (service / drug / mileage) to label each row.

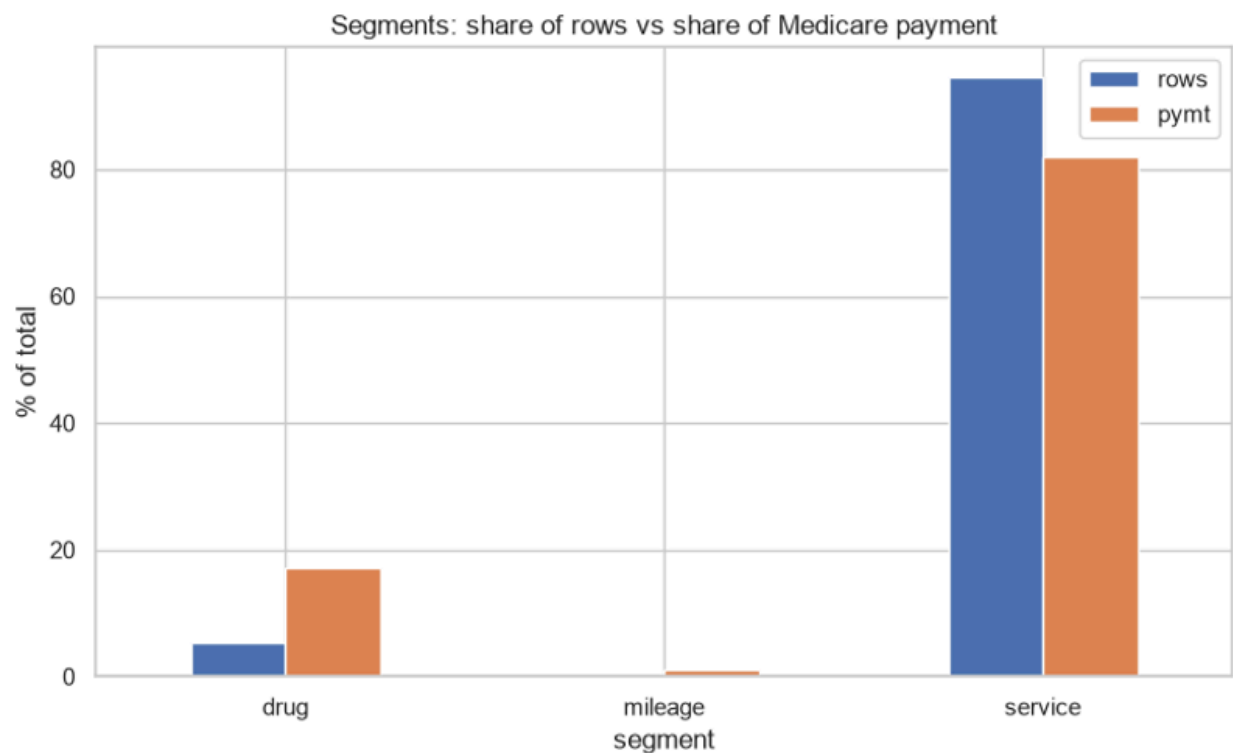


Fig 4. Segments: share of rows vs. share of Medicare payment.

- Drugs make up 5% of rows but 17% of Medicare payments, about 3× more money per row than average.
- Mileage is tiny overall, but each row carries about 10× the average payment, because one row sums all miles of all trips billed by that provider.

Therefore Tot_Srvcs cannot be summed or ranked across codes with different units (100 office visits, 3,000 mg of a drug and 400 miles are not comparable). In the model this is

handled by peer groups: since HCPCS is part of the peer group, drug units are only compared with the same drug, and miles with miles.

4.5 Cleaning summary

- Removed 422 rows with foreign addresses (Germany, Canada, Italy, Japan, etc.). U.S. territories (PR, GU, VI, AS) and military addresses (AA, AE, AP) were kept.
- Removed 90 rows that failed logic checks.
- Dropped columns not needed for analysis: street address, middle initial, FIPS code, country.
- Kept **9,781,161 of 9,781,673 rows (99.995%)** - the data is very clean.

Only 0.005% of rows were removed, so the cleaning does not change the statistics and figures below.

5. Descriptive Statistics

	mean	std	min	1%	25%	50%	75%	99%	max
Tot_Benes	84.99	1312.21	11.0	11.00	17.00	32.00	72.00	601.00	896762.00
Tot_Srvcs	281.83	6308.07	11.0	11.00	21.00	43.00	119.00	2753.00	6486401.00
Tot_Bene_Day_Srvcs	137.26	2181.50	11.0	11.00	20.00	41.00	106.00	1180.00	1835261.00
Avg_Sbmtd_Chrg	430.76	1521.62	0.0	5.00	73.28	180.00	367.00	4293.00	99999.99
Avg_Mdcr_Alowd_Amt	106.37	333.00	0.0	0.69	25.49	71.55	121.14	826.01	53965.43
Avg_Mdcr_Pymt_Amt	83.01	265.62	0.0	0.52	19.65	52.46	90.91	656.38	42994.89
Avg_Mdcr_Stdzd_Amt	82.50	267.16	0.0	0.53	19.47	52.66	90.78	652.10	42994.89

Table 3. Descriptive statistics of numeric variables (cleaned data, 9,781,161 rows).

- All variables are strongly right-skewed: the mean is much larger than the median (e.g. Tot_Srvcs: mean 282 vs. median 43), and the max is thousands of times larger than the 99th percentile.
- A typical service is charged at about \$180 (median), while Medicare allows about \$72 and pays about \$52.

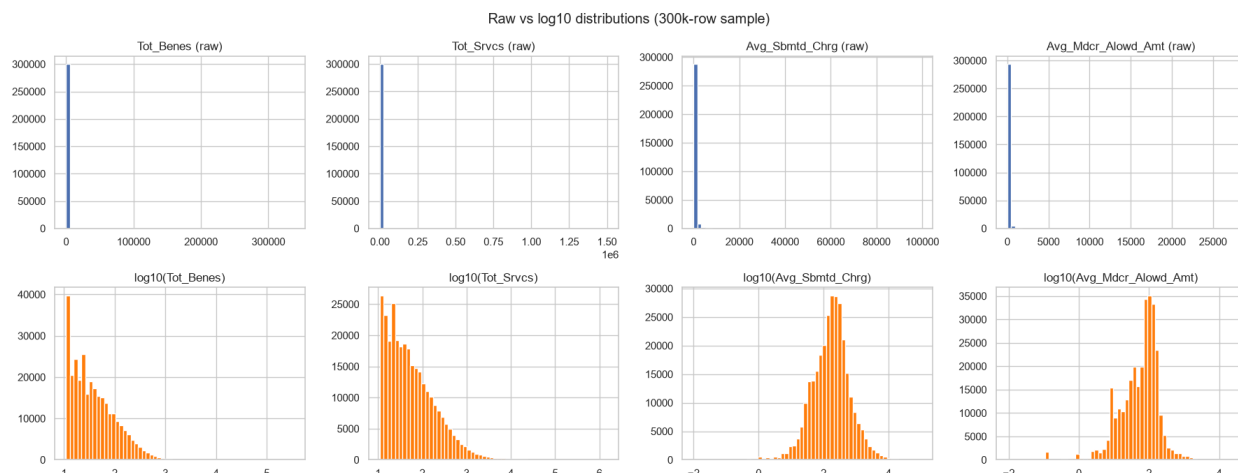


Fig 5. Skewed distributions: raw scale (top) vs. log10 scale (bottom).

On the raw scale almost all observations fall into a single histogram bar. After a log10 transformation the distributions become readable. Therefore:

- Variables must be log-transformed before modeling. Isolation Forest isolates points with random splits between a feature's minimum and maximum. On a raw scale most splits land in the empty range between typical values and a few extreme ones, so the model would only "see" the largest providers.
- Medians and MAD (median absolute deviation) should be used instead of means and standard deviations for peer comparisons. Extreme values inflate the mean and standard deviation and can hide the outliers themselves.

6. Exploratory Analysis & Visualizations

6.1 Provider types: records vs. unique providers

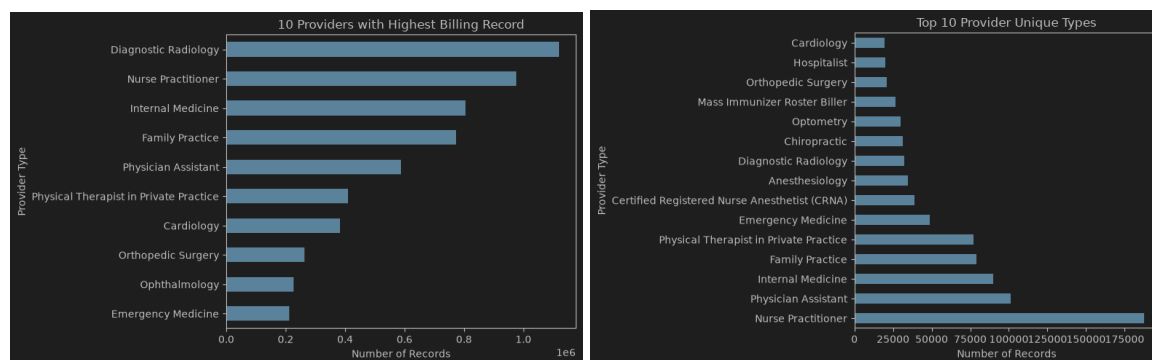


Fig 6. Provider types with the most provider-service records (left). Fig 7. Provider types with the most unique providers (right).

Because of how the data set is structured (every row is not its own provider, but rather each service listed for each provider, it is easy to confuse these two statistics.

Nurses being the most appearing unique provider type does not mean they have the most contribution. Diagnostic Radiology had the largest number of total CMS records, while Nurse Practitioners had the largest number of unique providers. The records-per-provider analysis helps explain this difference: some provider types, such as Clinical Laboratories and Diagnostic Radiology, contribute many more service records per individual provider, than nurse practitioners.

Provider type	Records per provider
Clinical Laboratory	39.7
Diagnostic Radiology	34.9
Portable X-Ray Supplier	28.7
Clinical Cardiac Electrophysiology	27.1
Micrographic Dermatologic Surgery	24.7
Interventional Radiology	24.5
Interventional Cardiology	22.4
Peripheral Vascular Disease	20.0
Cardiology	19.6
Urology	18.2

Table 4. Average number of provider-service records per unique provider (top 10 provider types).

6.2 Most common services (HCPCS)

The next important thing to look at is the most common service types and their patterns.

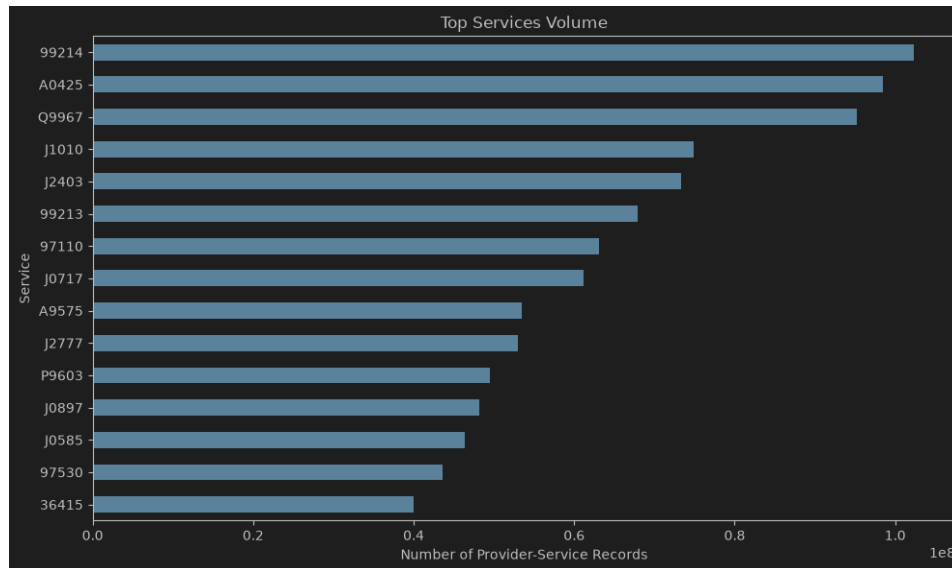


Fig 8. HCPCS codes with the most provider-service records.

	HCPCS_Cd	HCPCS_Desc	record_count	unique_providers	total_services	services_per_record
5463	99214	Established patient office or other outpatient...	523486	498045	102377857.0	195.569427
5462	99213	Established patient office or other outpatient...	473425	457569	67927936.1	143.481937
5458	99204	New patient office or other outpatient visit w...	239126	232660	11508274.0	48.126402
5469	99232	Subsequent hospital care with moderate levelof...	176550	176156	33827454.0	191.602685
5464	99215	Established patient office or other outpatient...	161945	153625	11827816.0	73.036006
5457	99203	New patient office or other outpatient visit w...	158518	155883	7685964.0	48.486380
5832	G2211	Visit complexity inherent to evaluation and ma...	146024	138770	24627539.1	168.654051
5467	99223	Initial hospital care with moderate level of m...	141311	140980	9607216.1	67.986329
5470	99233	Subsequent hospital care with moderate levelof...	131828	131573	22309090.0	169.228768
5466	99222	Initial hospital care with straightforward or ...	116128	115957	5366122.0	46.208684

Table 5. Top HCPCS codes: number of records, unique providers, total services and services per record

The most frequent HCPCS codes are mostly office/outpatient and hospital evaluation visits. Code 99214 appears in 523,486 records from 498,045 unique providers and represents more than 102 million services. Service intensity differs a lot between codes: some average nearly 200 services per record, others fewer than 50. So record frequency alone is not enough to compare services, which supports HCPCS-specific peer groups.

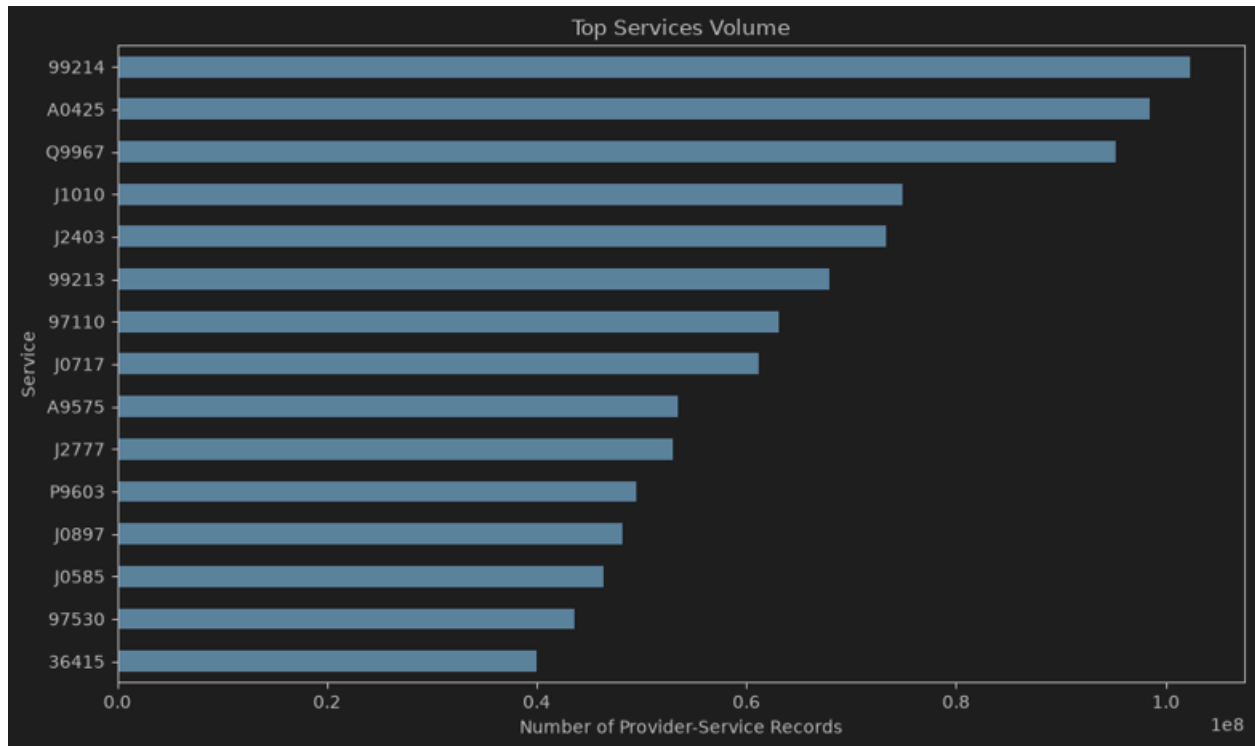


Fig 9. HCPCS codes with the highest total service volume (Tot_Srvcs).

Note: this ranking mixes units (Section 4.4). A0425 and P9603 count miles, and J/Q codes count drug units, so their totals are not comparable with visit codes like 99214.

6.3 Service intensity by provider type

Another important thing to note is that median service per patient by provider can differ. For some practice types 1 will be the norm, while for others it will be 5. But it does not mean that the 5 is an outlier, it just simply belongs to a different category.

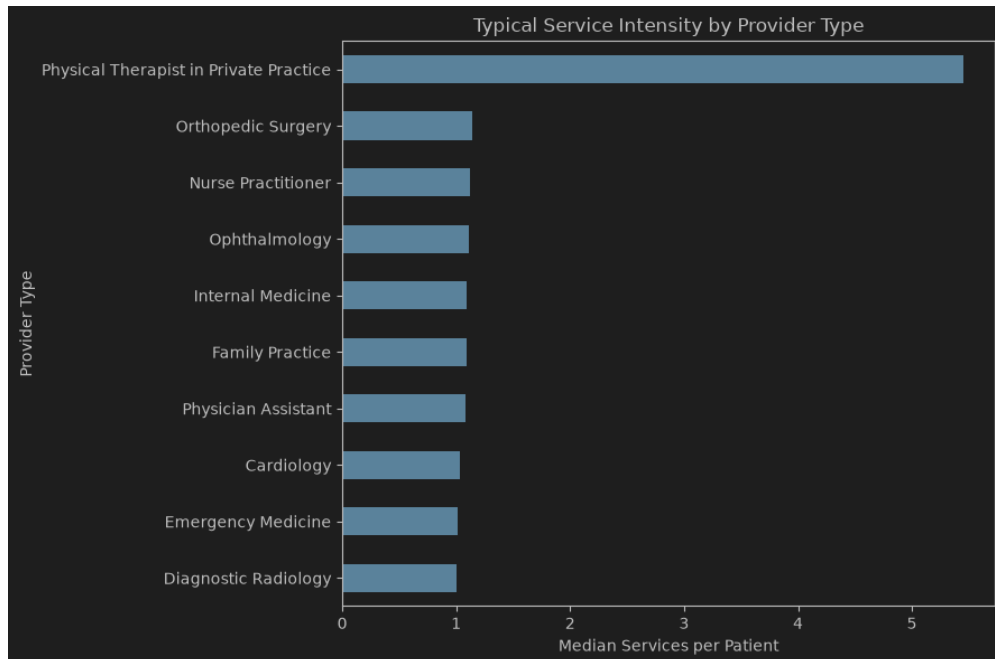


Fig 10. Median services per patient by provider type.

6.4 Place of service

Place_Of_Srvc	F	O
Rndrng_Privr_Type		
Diagnostic Radiology	71.500585	28.499415
Nurse Practitioner	22.355088	77.644912
Internal Medicine	29.918183	70.081817
Family Practice	9.830440	90.169560
Physician Assistant	23.588708	76.411292
Physical Therapist in Private Practice	0.038165	99.961835
Cardiology	47.461591	52.538409
Orthopedic Surgery	24.220123	75.779877
Ophthalmology	16.023744	83.976256
Emergency Medicine	83.325434	16.674566

Table 6. Share of records in facility (F) vs. office (O) settings, % (top 10 provider types).

Place-of-service distribution varies substantially across provider types. Emergency Medicine and Diagnostic Radiology are predominantly represented in facility settings, while Family Practice and Physical Therapists in Private Practice are overwhelmingly non-facility based. Cardiology shows a more balanced distribution. Place of service is strongly related to specialty, so it should be part of the peer group.

6.5 Billing markup (charge/allowed amount)

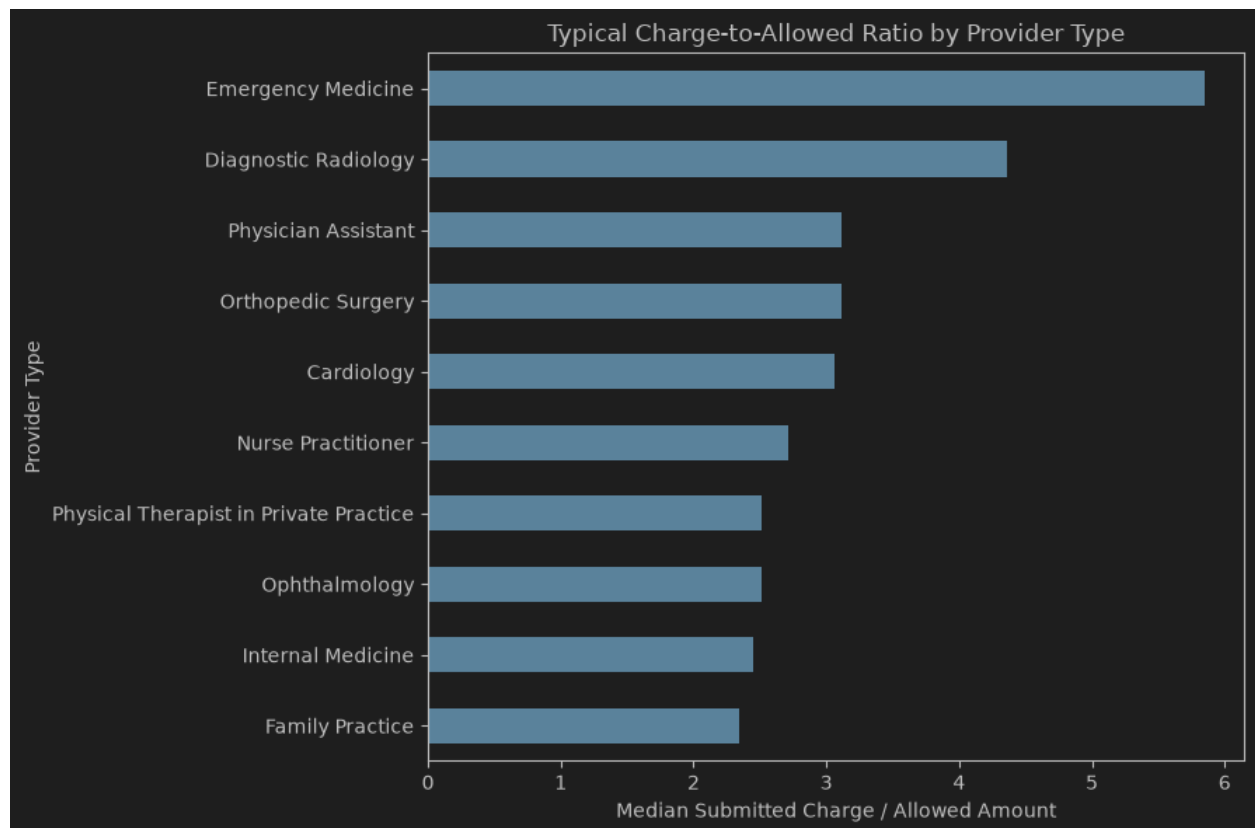


Fig 11. Median submitted charge / allowed amount by provider type.

Emergency Medicine and Diagnostic Radiology have the highest typical ratios, while Family Practice and Internal Medicine have the lowest. Billing intensity depends on specialty, so charge-based anomaly detection should compare providers against similar peers rather than use one threshold for everyone.

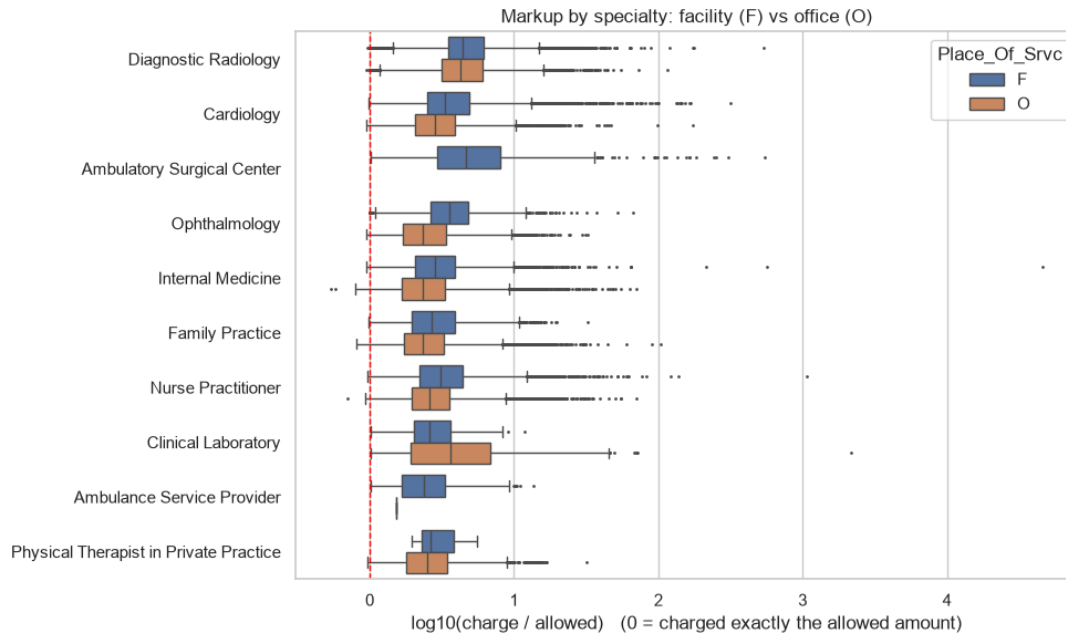


Fig 12. Charge ÷ allowed by specialty, split by facility (F) and office (O). The red line means "charged exactly the allowed amount".

- Providers usually charge 2.5–4× more than Medicare allows.
- In most specialties the markup is higher in facilities (e.g. Ophthalmology: 3.5× vs 2.4× in offices), because Medicare pays doctors less there.

Therefore facility and office rows have different "normal" values, so they must be compared separately - peer groups = specialty × HCPCS × place of service.

We calculated average submitted charges for providers within each provider type and created charts showing the 100 highest averages. Most have a few firms that are the highest charging and it slowly decreases, however there are a few providers that are showing irregular distributions. For example among the Chiropractic providers, there is one provider that charges over 10 times higher than the second highest charging provider. Another example is Home Infusion Therapy Services where the highest charging provider charges over 15 times higher than the second highest charging provider. These charts show only the top 100 highest averages within each provider type, so they do not represent the whole distribution. However, there is a big enough difference to suggest possible unusual behaviour among these providers. Averages mix different HCPCS codes, so these differences may reflect service mix rather than unusual billing; this will be checked within peer groups.

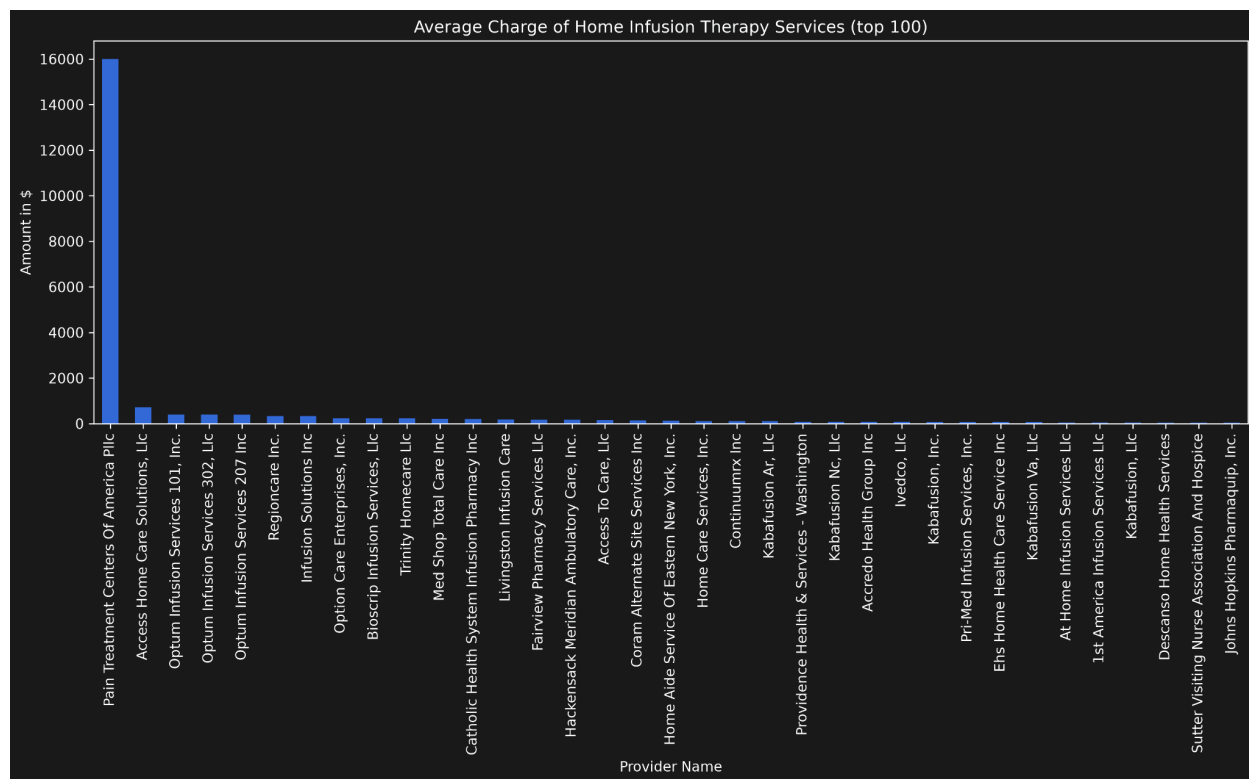
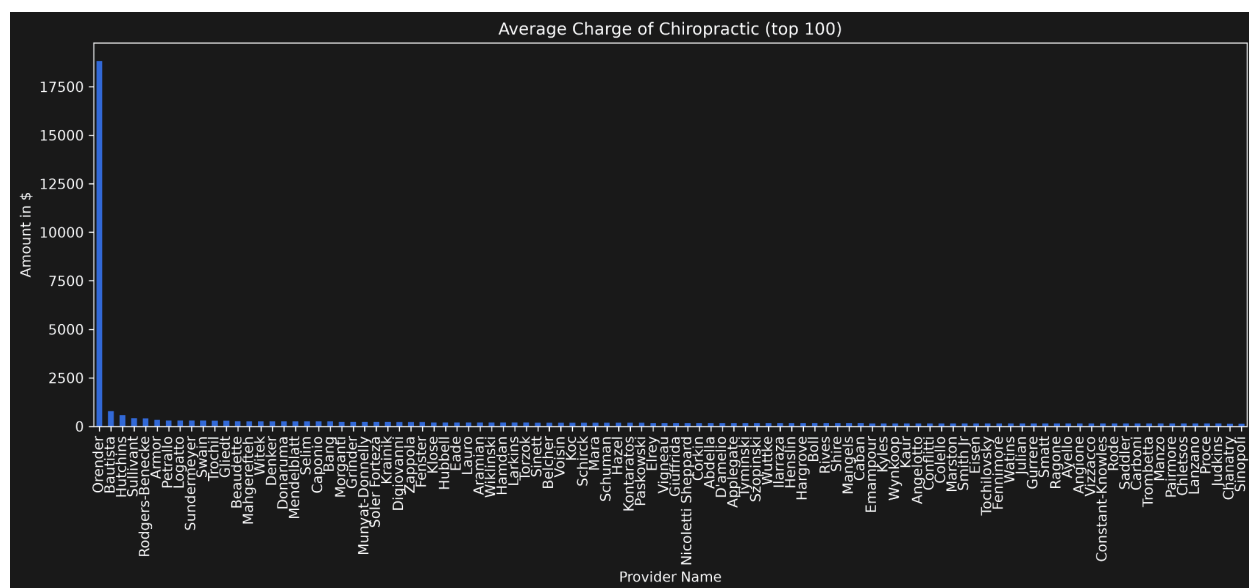


Fig 13. Provider names with Average charge.

6.6 Geography

Rank	State	Unique providers	State	Total services (M)
1	CA	96,367	FL	302.2
2	NY	82,431	CA	278.1
3	TX	80,776	TX	214.4
4	FL	80,193	NY	166.6
5	PA	59,281	NJ	133.3
6	OH	48,707	PA	115.5
7	IL	48,565	IL	110.1
8	NC	41,621	NC	90.3
9	MI	39,713	MD	88.7
10	NJ	37,851	VA	80.2

Table 7. Top 10 states by number of unique providers and by total reported services.

Many providers does not necessarily mean many services. California has the most unique providers, while Florida has the highest total service volume. Raw state totals should be read with caution: states differ in population, Medicare enrollment and provider mix, and total services mix different units (Section 4.4).

We counted unique providers within each state by provider type and created bar charts to compare their distributions. While most states have Nurse Practitioners as the highest number of providers, there are some states that don't. For example California has the highest number of providers being Internal Medicine. Another example is Hawaii where it's Physical Therapist in Private Practice. These results show that the most common provider types differ among states.

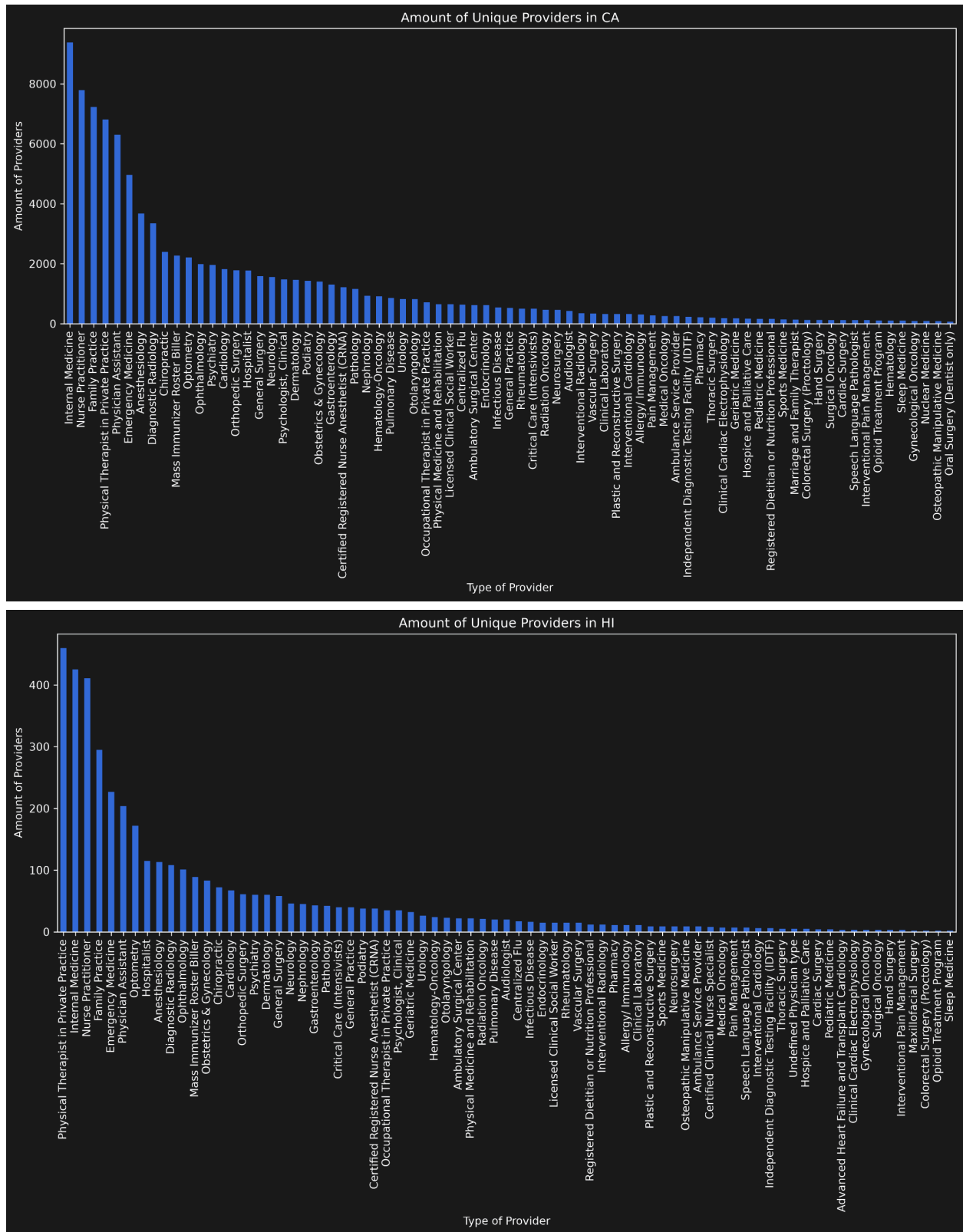


Fig 14. Amount of Unique Providers in States.

6.7 Part B drugs

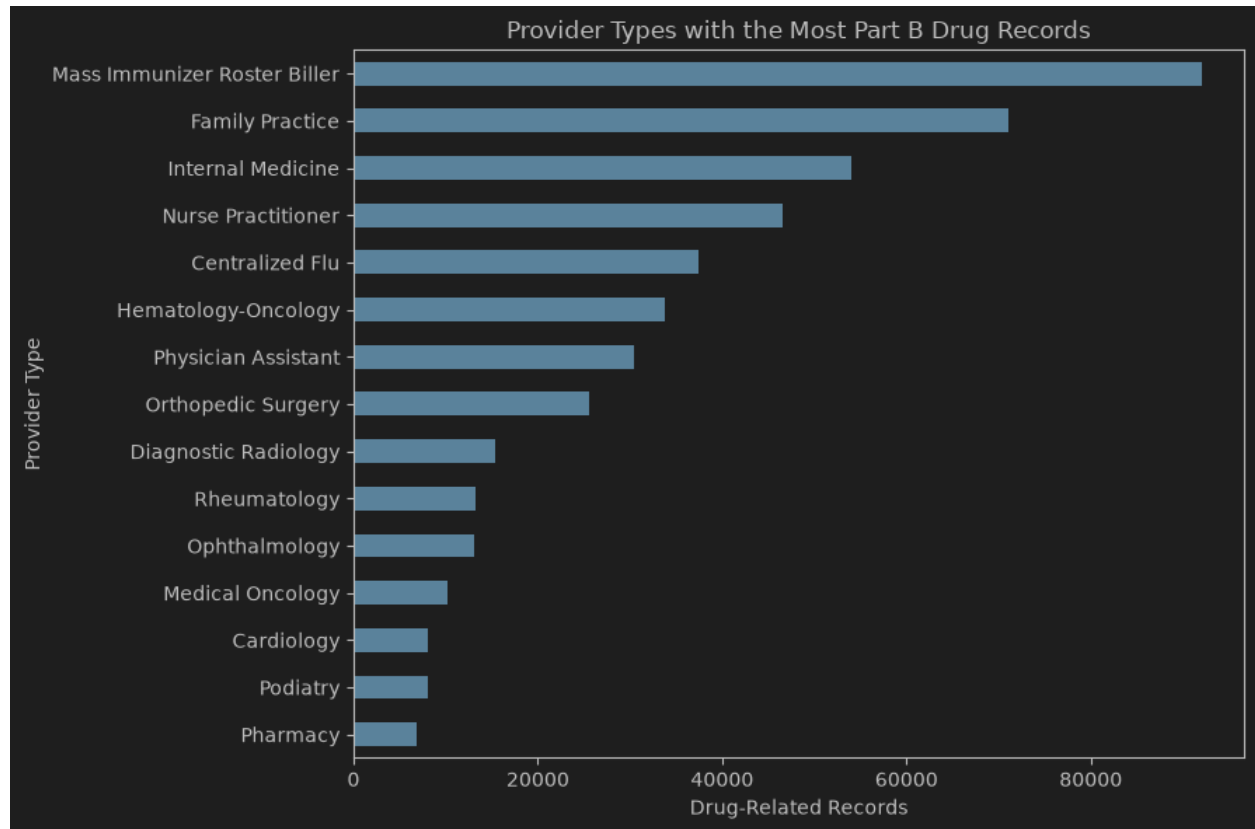


Fig 15. Provider types with the most Part B drug records.

Part B drug records are concentrated in a small set of provider types. Mass Immunizer Roster Billers have the most drug records, followed by primary-care types such as Family Practice, Internal Medicine and Nurse Practitioners.

6.8 Correlations between numeric variables

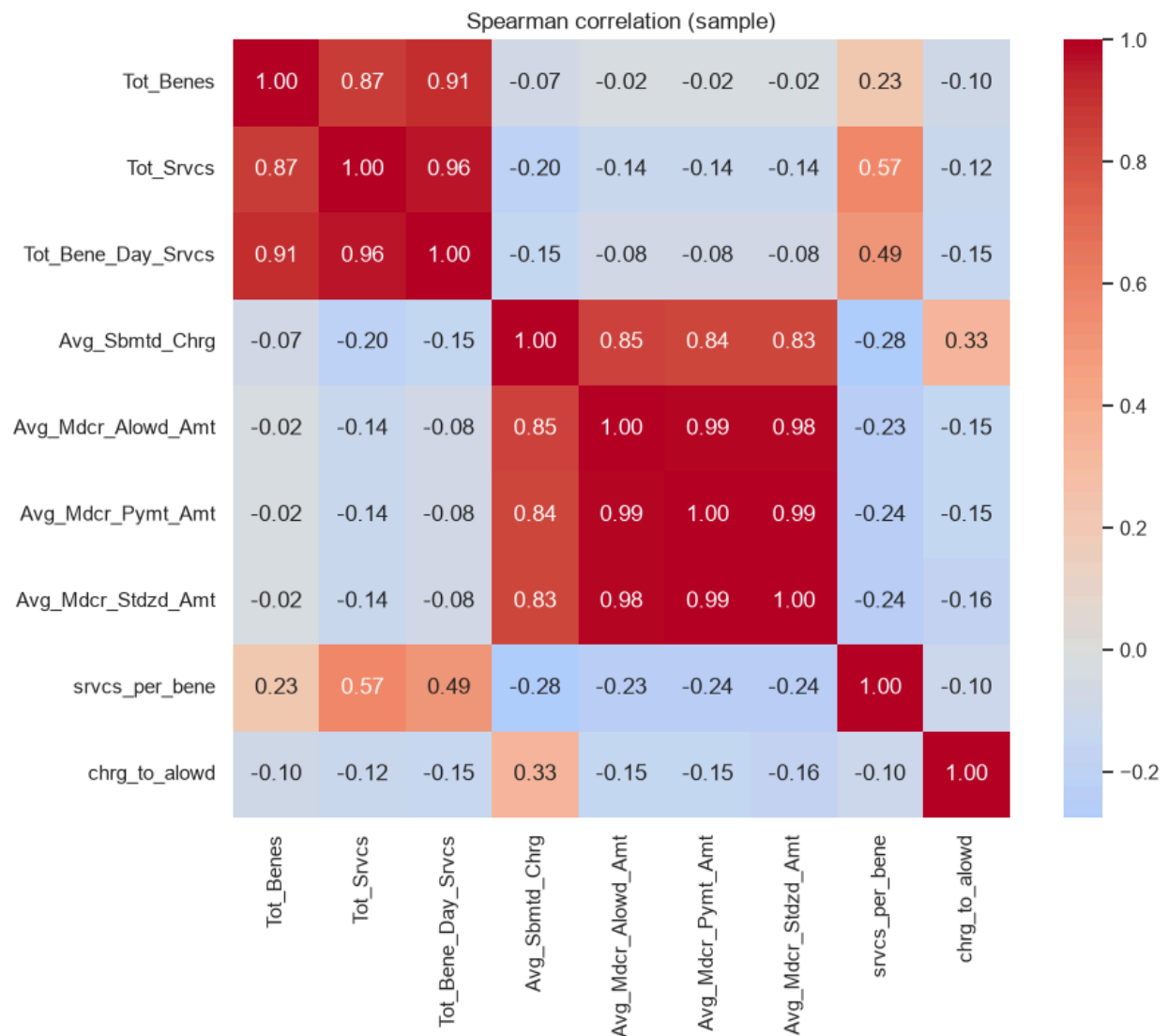


Fig 16. Spearman correlation between volume, payment and ratio variables.

- Two groups of columns are almost duplicates:
- Volume: patients / services / patient-days (0.87–0.96)
- Money: allowed / payment / standardized payment (0.98–0.99)
- Ratios (services per patient, charge/allowed) are not strongly related to the raw columns, so they add new information.

Instead of all 7 raw columns, the model could use one column per group plus ratios: patients, services per patient, patient-days per patient, markup, and standardized payment.

Isolation Forest does not break with correlated features, but duplicates get more weight in random splits. We suggest using the reduced set by default and compare it with the full set in the sensitivity analysis.

6.9 Additional Exploratory Analysis and Interpretation

1) Peer-Group Fallback usage

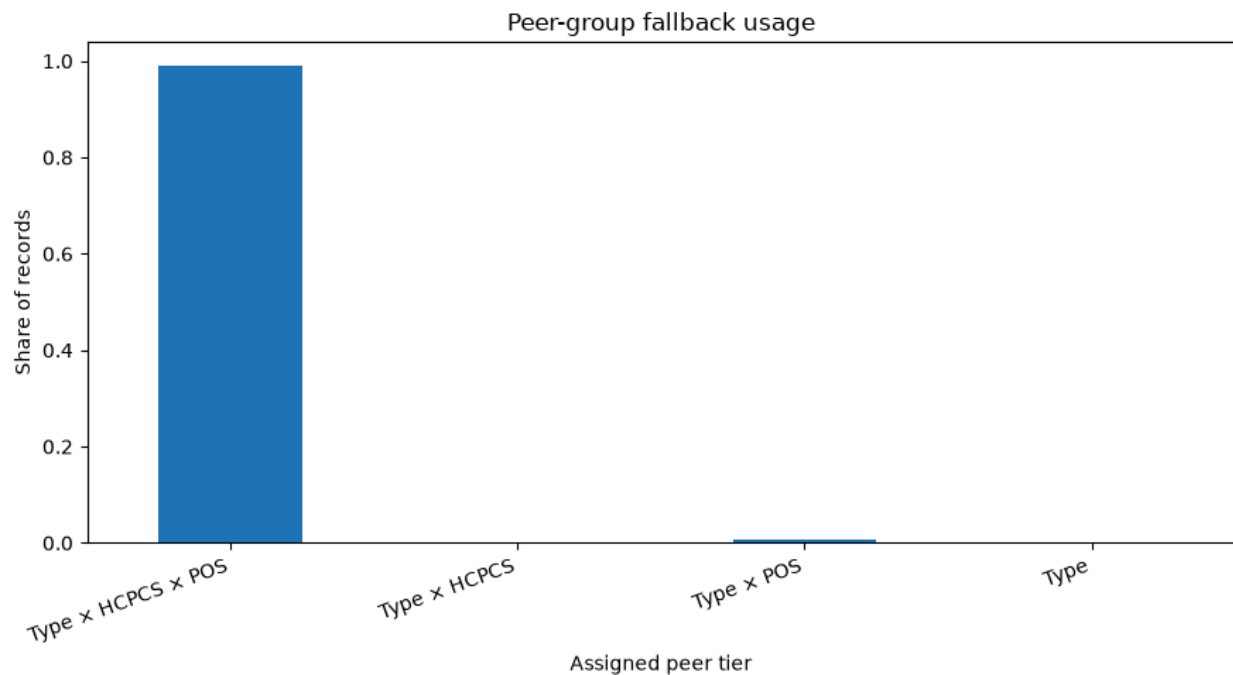


Fig 17. Grouped features, with fallbacks if observations <10.

What the graph shows: 98.96% of records are assigned to the most specific peer group: provider type × HCPCS × place of service. About 0.192% fall back to provider type × HCPCS, 0.846% to provider type × place of service, and about 0.0003% to provider type alone.

Why this matters: The intended peer group is usable for nearly the entire dataset. The fallback tiers provide a way to retain observations whose most-specific group is too small for a reliable comparison. The notebook therefore supports the report's planned peer-group structure rather than comparing all providers against one overall population.

2) Provider behavior features

Two provider-level descriptive features from the notebook: E&M service mix and HCPCS concentration (HHI). These capture service mix and concentration that raw volume alone does not capture.

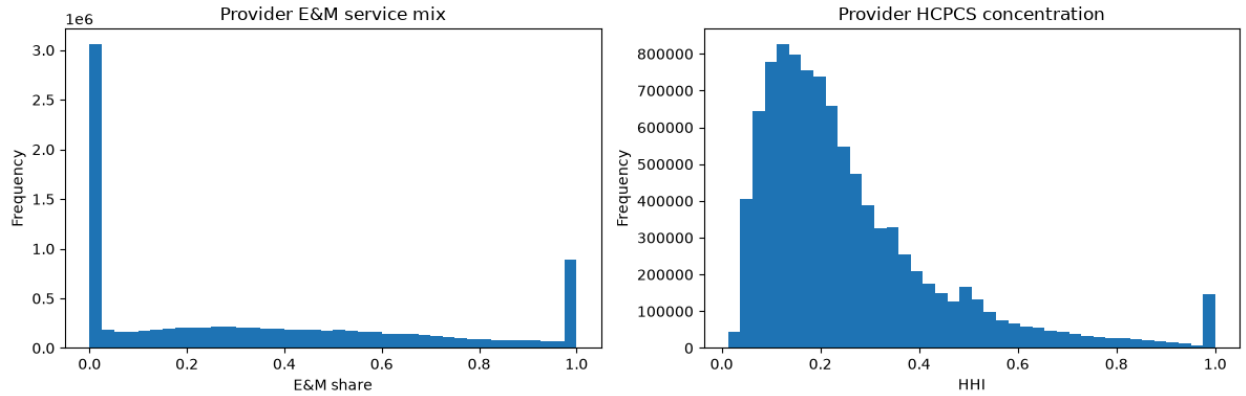


Fig 18 (left). Share of provider's services that are E&M

Fig 19 (Right). Concentration of a providers activity across HCPCS codes

	em_mix	hcpcs_hhi	hcpcs_count
count	1207388	1207388	1207388
mean	0.462	0.416	7.80
25%	0.000	0.213	3.00
media n	0.452	0.333	5.00
75%	0.919	0.523	9.00
95%	1.000	1.00	24.00
max	1.000	1.000	659.00

Table 8. (em_mix) large concentrations near 0 and 1, meaning little to no E&M activity.

(hhi) A value closer to 1 means activity is concentrated with few codes.

E&M mix measures the share of a provider's service units that are E&M services. HCPCS HHI measures how concentrated the provider's activity is across HCPCS codes; values closer to 1 indicate concentration in fewer codes. These are descriptive behavioral features, not fraud labels or proof of improper billing

3) Which variables help define peer groups?

The notebook calculated eta-squared (η^2) for log standardized Medicare payment. This measures how much variation in the payment variable is associated with differences between groups defined by each candidate variable

	candidate	group s	eta_squar ed
1	HCPCS_Cd	6471	0.920633
0	Rndrng_Prldr_Type	113	0.151156
6	HCPCS_Drug_Ind	2	0.024512
2	Place_Of_Srvc	2	0.022096
4	Rndrng_Prldr_Ent_Cd	2	0.008265
5	Rndrng_Prldr_State_A brvtn	61	0.004356
3	Rndrng_Prldr_RUCA	23	0.001947

Table 9. Eta-squared table measures how much variation in the analyzed standardized payment outcome is associated with differences between the groups.

HCPCS has by far the largest association with standardized payment variation in this analysis, followed by provider type. Place of service explains little of the payment level, but it strongly affects markup (Fig 12) and specialty mix (Table 6), so it is kept in the peer group. State and RUCA have much smaller η^2 values here, so the notebook supports keeping geography primarily as contextual information while using provider type $\times$ HCPCS $\times$ place of service as the core peer-group definition.

7. Target Variable and Key Features

This project uses unsupervised anomaly detection: there is no target variable such as Fraud = yes/no in this dataset. Instead, the model will use the following features:

Peer-group variables: Rndrng_Prldr_Type, HCPCS_Cd, Place_Of_Srvc

Model features (reduced set): log(Tot_Benes), services per patient (Tot_Srvcs / Tot_Benes), patient-days per patient (Tot_Bene_Day_Srvcs / Tot_Benes), markup (Avg_Sbmtld_Chrg / Avg_Mdcr_Alowd_Amt), log(Avg_Mdcr_Stdzd_Amt)

Full set for comparison: all 7 raw volume and payment columns (log-transformed)

Context / dashboard variables: state, RUCA, entity type, segment (service / drug / mileage), HCPCS_Drug_Ind

8. Initial Findings

The data is very clean. 99.995% of rows were kept; missing values are structural and appear only in text columns.

Records $\neq$ providers. Diagnostic Radiology contributes the most records, while Nurse Practitioners are the most common provider type.

Units differ. Tot_Srvcs counts visits, drug units or miles depending on the code, so volumes are only comparable within the same HCPCS code. Code 99214 is the most common service (498,045 providers, 102M+ services).

Normal behavior depends on specialty and place of service. Services per patient, facility vs. office share and markup all vary by provider type, and markup is higher in facilities.

Geography. California has the most providers, Florida the highest service volume; raw state totals need normalization.

Distributions are extremely right-skewed, so log transformation and robust statistics (median/MAD) are needed.

Volume and payment columns are highly correlated ($\rho = 0.87\text{--}0.99$), so a reduced feature set is enough.

9. Planned Next Steps

Build peer groups (provider type $\times$ HCPCS $\times$ place of service) and use fallback groups for peer groups with < 10 observations for reliable comparison. Also, test variations of different factors (grouped features, minimum group size, etc.)

Engineer ratio features and apply log transformations.

Compute a baseline: robust z-scores (median/MAD) and percentile flags within each peer group.

Train an Isolation Forest on the reduced feature set and compare it with the full feature set (sensitivity analysis).

Compare flags from the baseline and Isolation Forest and aggregate scores to the provider level.

Build the Power BI dashboard (filters by state, specialty and HCPCS; provider drill-down).

10. Team Member Contributions (Self-Declaration)

Diana: column descriptions; missing-value and duplicate checks; provider-type analysis (records vs. unique providers, records per provider); HCPCS frequency and volume analysis; service intensity and place-of-service analysis; geographic analysis; Part B drug analysis; charge-to-allowed ratio by provider type; key features and initial findings.

Alisa: data loading and initial inspection; logical consistency checks and unique-key validation; Tot_Srvcs unit investigation (drug units, mileage) and segment column; data cleaning; descriptive statistics and skewness / log-transformation analysis; facility vs. office markup comparison; Spearman correlation and reduced feature set; report integration.

Erica: GitHub documentation/organization/version control, report integration into git, data preprocessing/understanding, feature engineering for isoforest, correlation matrix analysis, aggregation of Part A report; Human readability of code and models.

Tyson: counted unique providers in each state and analyzed their distribution by provider type; calculated average submitted charges by provider within each provider type; created state-level provider-count charts and charts showing the top 100 providers by average submitted charge within each provider type.

Eli: Additional exploratory analysis and interpretation: initial observations into peer grouping, provider behavior choices, and critical features to define peer groups)